

// Core Logic //

if (stop > k):

continue;

else:

for each (neighbour, price) in graph[node]:

if (curr_cost + price < cost[neighbour]):

cost[neighbour] = curr_cost + price

push(stops + 1, neighbour, cost[neighbour])

if

// Complexity Analysis //

Time Complexity:

• Building adjlist $\rightarrow O(E)$ edges

• BFS traversal $\rightarrow O(E)$

Total time complexity $\Rightarrow \underline{\underline{O(E)}}$

Space Complexity:

• Adjlist $\rightarrow O(E)$

• Cost array $\rightarrow O(V)$

• Queue $\rightarrow O(V)$

Total space complexity $\rightarrow \underline{\underline{O(V+E)}}$